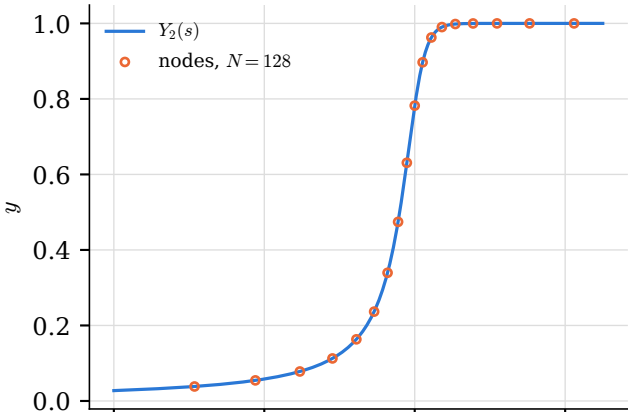
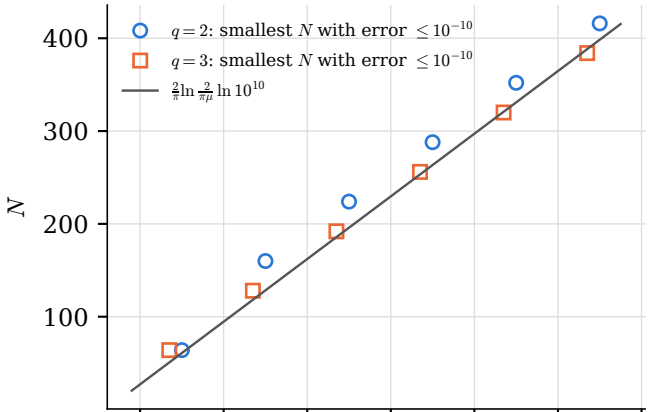

(a) $q = 2$, $\delta = 10^{-6}$: nodes near the front
$Y_2(s)$
nodes, $N = 128$
1.0
0.8
0.6
0.4
0.2
0.0
y
−40
−20
0
20
$s = (x - x_0)/\mu$


(b) nodes needed for 10^{-10}
$q = 2$: smallest N with error $\le 10^{-10}$
$q = 3$: smallest N with error $\le 10^{-10}$
$\frac{2}{\pi}\ln\frac{2}{\pi\mu}\ln 10^{10}$
400
300
200
100
N
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
layer width parameter μ